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Question 1

To prove that $2^{70} + 3^{70}$ is divisible by 13, we show that:

$$2^{70} + 3^{70} \equiv 0 \pmod{13}$$

Now, we compute each term modulo 13 separately.

- $2^{70} \pmod{13}$

Since $N = 13$ is a prime number, we use Euler's totient function:

$$\phi(13) = 12$$

By Euler's theorem:

$$2^{12} \equiv 1 \pmod{13}$$

We rewrite 2^{70} as:

$$(2^{12})^5 \times 2^{10} \pmod{13}$$

Since $(2^{12})^5 \equiv 1^5 \equiv 1 \pmod{13}$, we get:

$$2^{70} \equiv 2^{10} \pmod{13}$$

Using calculations, $2^{10} \equiv 10 \pmod{13}$, so:

$$2^{70} \equiv 10 \pmod{13}$$

- $3^{70} \pmod{13}$

Similarly, using Euler's theorem:

$$3^{12} \equiv 1 \pmod{13}$$

We rewrite 3^{70} as:

$$(3^{12})^5 \times 3^{10} \pmod{13}$$

Since $(3^{12})^5 \equiv 1^5 \equiv 1 \pmod{13}$, we get:

$$3^{70} \equiv 3^{10} \pmod{13}$$

Using calculations, $3^{10} \equiv 3 \pmod{13}$, so:

$$3^{70} \equiv 3 \pmod{13}$$

Now, compute the sum modulo 13:

$$2^{70} + 3^{70} \equiv 10 + 3 \pmod{13}$$

Thus, we conclude:

$$2^{70} + 3^{70} \equiv 0 \pmod{13}$$

Then 13 divides $2^{70} + 3^{70}$

Question 2

For any natural numbers a and b , let's denote:

$$x = 2^a - 1, \quad y = 2^b - 1$$

To find $\gcd(x, y)$, we use the fact that:

$$\gcd(2^m - 1, 2^n - 1) = 2^{\gcd(m, n)} - 1$$

Therefore:

$$\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a, b)} - 1$$

Proof:

Express the given numbers in terms of powers of 2. Let $d = \gcd(a, b)$. Then there exist integers k and m , $\gcd(k, m) = 1$ such that:

$$a = kd, \quad b = md$$

Thus:

$$2^a - 1 = 2^{kd} - 1, \quad 2^b - 1 = 2^{md} - 1$$

Using the property of GCD for numbers of the form $2^x - 1$:

$$\gcd(2^x - 1, 2^y - 1) = 2^{\gcd(x, y)} - 1$$

Applying this to our case:

$$\gcd(2^{kd} - 1, 2^{md} - 1) = 2^{\gcd(kd, md)} - 1$$

Since:

$$\gcd(kd, md) = d \cdot \gcd(k, m)$$

and because k and m are coprime, we get:

$$\gcd(kd, md) = d$$

Thus, substituting back:

$$\gcd(2^{kd} - 1, 2^{md} - 1) = 2^d - 1$$

Therefore, we conclude:

$$\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$$

Question 3

Evaluate $\gcd(9n + 4, 2n1)$

Using Extended Euclidean Algorithm

$$\begin{aligned} 9n + 4 &= (2n - 1) \cdot 4 + (n + 8) \\ 2n - 1 &= (n + 8) \cdot 1 + (n - 9) \\ n + 8 &= (n - 9) \cdot 1 + 17 \end{aligned}$$

So,

$$(n + 8) \equiv 17 \pmod{n - 9} = n + 8 \pmod{n - 9} = 17$$

Thus, the gcd simplifies further:

$$\gcd(n + 8, n - 9) = \gcd(n - 9, 17)$$

Since 17 is a prime number, $\gcd(n - 9, 17)$ is either 1 or 17, depending on whether $n - 9$ is divisible by 17.

- If $n - 9$ is not a multiple of 17, then $\gcd(n - 9, 17) = 1$.
- If $n - 9$ is a multiple of 17, then $\gcd(n - 9, 17) = 17$.

1 Question 4

By contradiction, assume that n is not a power of 2. Let us consider $n = M \cdot 2^k$, where M is an odd prime number.

For any odd number M , we can write the expression:

$$x^M + 1 = (x + 1)(1 - x + x^2 + \cdots + x^{M-1}).$$

Now, we compute the summation:

$$1 - x + x^2 + \cdots + (-1)^{M-1}x^{M-1} = \frac{a(r^M - 1)}{r - 1}$$

where $a = 1$ and $r = -x$.

$$\begin{aligned}
1 - x + x^2 + \cdots + (-1)^{M-1}x^{M-1} &= \frac{(-x)^M - 1}{-x - 1} \\
&= \frac{(-1)^M x^M - 1}{-x - 1} \\
&= \frac{-x^M - 1}{-x - 1} \\
&= \frac{-(x^M + 1)}{-(x + 1)} \\
&= \frac{x^M + 1}{x + 1}.
\end{aligned}$$

Since M is odd, we know that $M - 1$ is even, so $(-1)^{M-1} = 1$. Since $n = M \cdot 2^k$, then:

$$\begin{aligned}
1 - x + x^2 + \cdots + x^{M-1} &= \frac{x^M + 1}{x + 1}, \\
(1 - x + x^2 + \cdots + x^{M-1})(x + 1) &= x^M + 1, \\
(1 - x + x^2 + \cdots + x^{M-1})(x + 1) &= (x^{2^k})^M + 1 = x^2 + 1.
\end{aligned}$$

Since $(x^{2^k})^M + 1$ is composite, this implies that $x^n + 1$ is not prime. so $2^2 + 1$ is prime then n is power of 2 .

Counterexample: Let $n = 2^5 = 32$. We compute:

$$2^{32} + 1 = 4294967297$$

Now, we check whether it is prime.

Step 1: Factorization

$$4294967297 = 641 \times 6700417$$

Since 4294967297 is composite, we have found a counterexample. Thus, the statement

”If n is a power of 2, then $2^n + 1$ is prime.”

is false.

2 Question 5

We are given:

- p is an **odd prime**.
- p does not divide a or b , i.e., $\gcd(a, p) = 1$ and $\gcd(b, p) = 1$.
- p divides $a^2 + b^2$, meaning:

$$a^2 + b^2 \equiv 0 \pmod{p}.$$

This can be rewritten as:

$$a^2 \equiv -b^2 \pmod{p}.$$

Step 2: Quadratic Residues Modulo p

The key idea in this proof involves quadratic residues modulo p .

- The **Legendre symbol** $\left(\frac{x}{p}\right)$ determines whether x is a quadratic residue modulo p .
- An important fact in number theory states that:

$$-1 \text{ is a quadratic residue modulo } p \text{ if and only if } p \equiv 1 \pmod{4}.$$

This means there exists some integer x such that:

$$x^2 \equiv -1 \pmod{p}$$

if and only if $p \equiv 1 \pmod{4}$.

Step 3: Applying This to Our Given Equation

Since we have:

$$a^2 \equiv -b^2 \pmod{p},$$

we divide both sides by b^2 (which is valid since p does not divide b):

$$\left(\frac{a}{b}\right)^2 \equiv -1 \pmod{p}.$$

This shows that -1 is a quadratic residue modulo p , which is only possible if:

$$p \equiv 1 \pmod{4}.$$

Thus, we have proven the statement. □